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The challenges of urban tunnelling

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ABSTRACT: The prime challenge of urban tunnelling involves control of ground stability and deformation as the governing design criterion, which must satisfy acceptable surface settlement limits. This presentation focuses on the present usage of settlement limits, on the progress achieved in development of theoretical models to predict the behaviour of a shallow tunnel, on the application of observational method of design, on the current state of ground conditioning and on the development of tunnel boring machines with positive face control.

1 CHALLENGES OF URBAN TUNNELLING

In general, any tunnelling project – urban or otherwise - requires three steps in preparation and design. These steps are:

a) Selection of a tunnelling system (a tunnelling method with or without additional measures including eventual ground conditioning) which is suitable for and compatible with the given ground;
b) Cost and risk analysis to determine the optimum choice for cases where more than one tunnelling system is applicable;
c) Detail design of tunnel which usually involves:
   1) estimation of loads on the liner
   2) structural design of the liner
   3) prediction of ground behaviour and deformations
   4) selection of measures to control ground behaviour.

While all the steps above are important, in the case of urban tunnels it is usually the ground behaviour and ground control which eventually become the governing criteria of design. This involves both the ground stability and ground deformation and, above all, as their conseqence – the surface settlement.

Reasons for this priority are most obvious, nevertheless it would be appropriate to reiterate them here. Urban environment is extremely sensitive to any type of infringement, interference and pollution. Many of these intervening factors are controlled by legislative regulations, others simply by public concern or political pressure. The fact is that the public has grown increasingly concerned about activities which may impact on their neighborhoods or their cities. Excavation of underground space is definitely one of those activities.

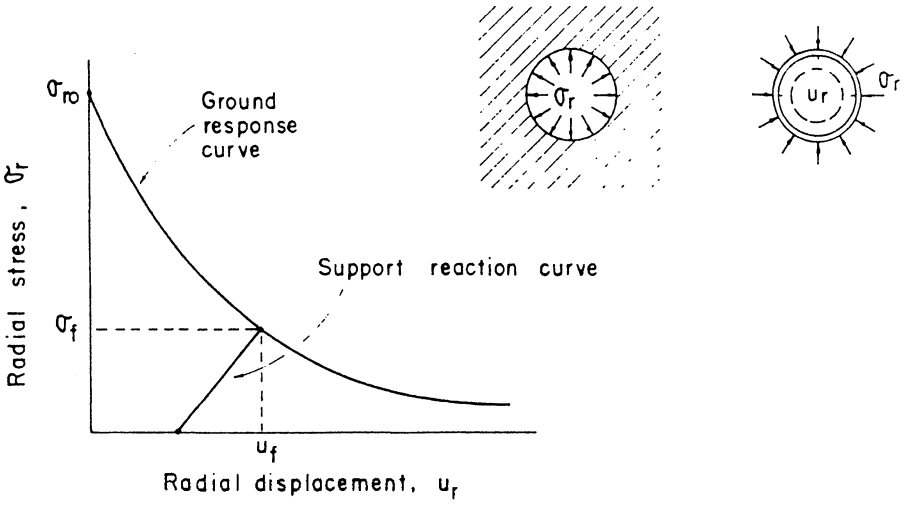


Figure 1. Convergence – confinement concept.

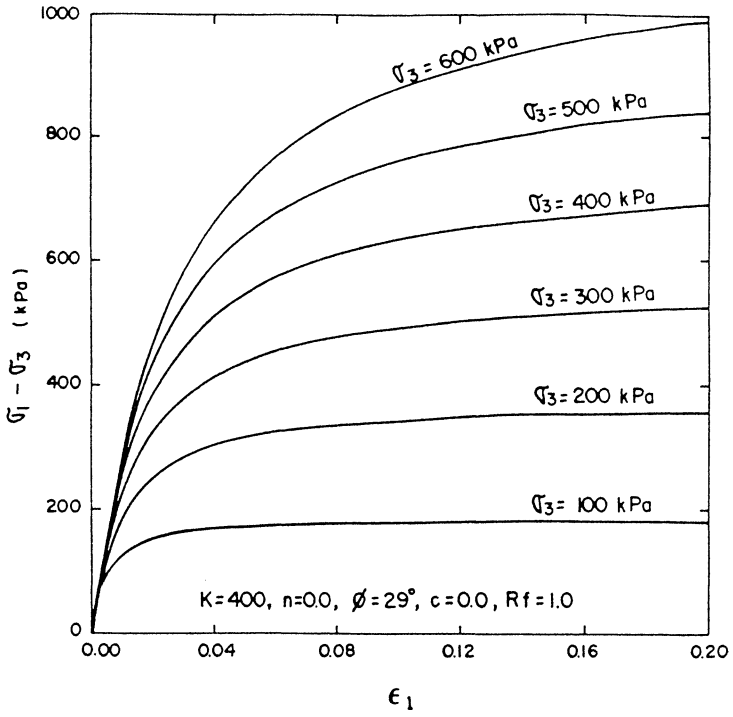


Figure 2. Homologous stress – strain hyperbolas.

2 PROGRESS OF URBAN TUNNELLING

The engineering of underground space has recognized this challenge for more than three decades. The past efforts can be roughly divided into the following categories:

a) Defining the limits of surface deformations according to technical, operational and social criteria;
b) Theoretical studies focused on the complex ground response to tunnelling and on the ground-structure interaction involved in excavation and support of underground space, on sources of loss of ground and on the mechanics of loss of ground distribution throughout the overburden and particularly on surface;
c) Development of the Observational Method as a way of design and construction control and as a warning system against unacceptable behaviour;
d) Practical development in ground conditioning such as various types of grouting, freezing and ground water lowering;
e) Development of tunnel boring machines with positive face control such as the slurry and earth pressure balance TBM.

In the following, this presentation discusses the progress achieved in each of the five categories above, with an emphasis on the theoretical studies.

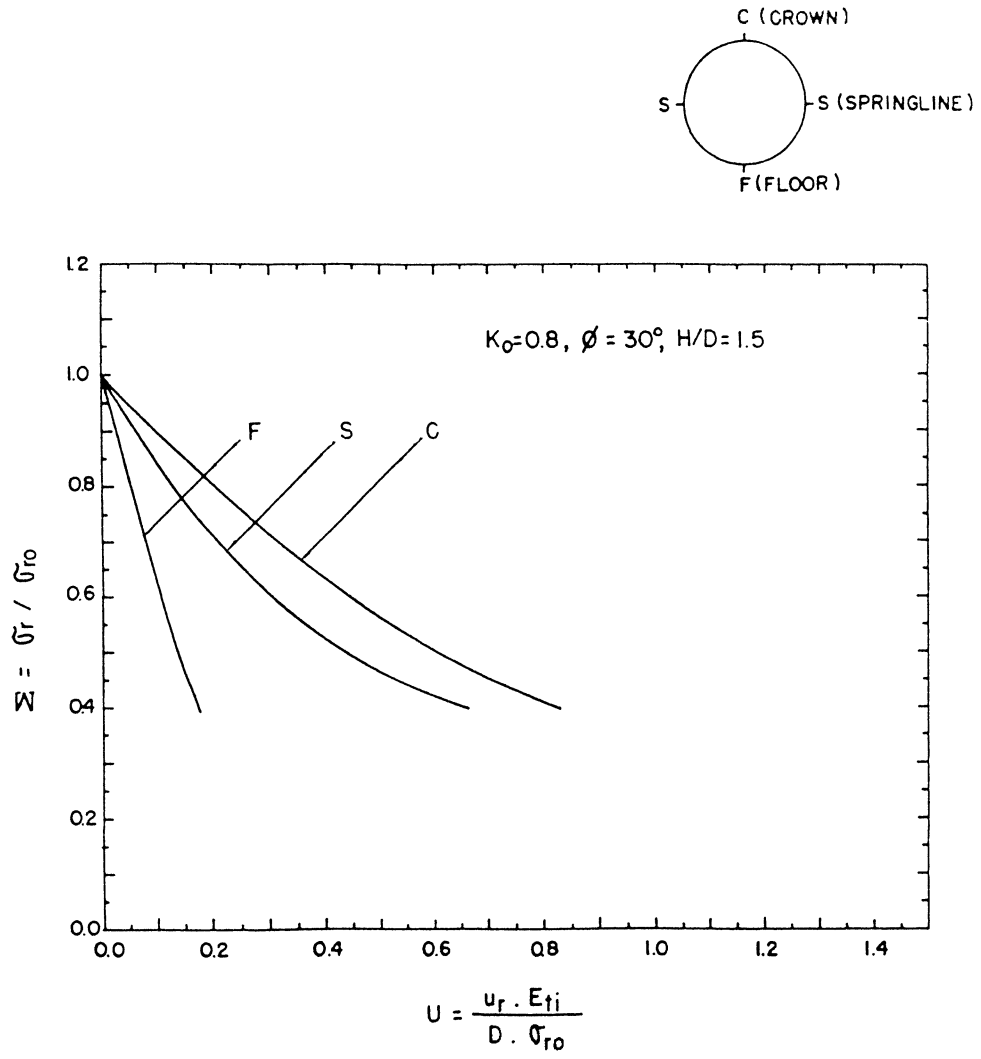


Figure 3. Typical normalized ground response curve.

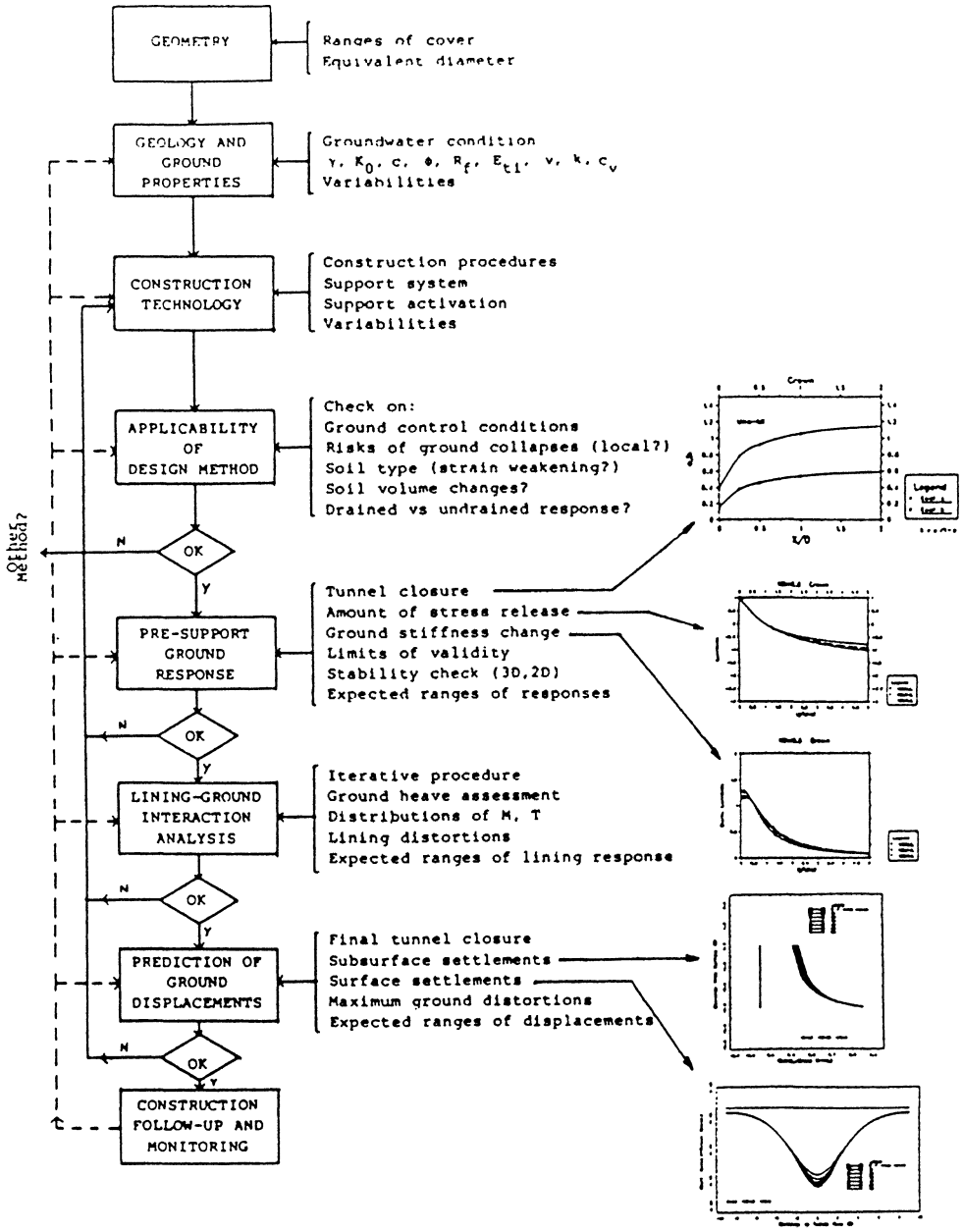


Figure 4. Suggested sequence of step for shallow tunnel design (after Negro 1988).

3 LIMITS OF SURFACE DEFORMATIONS

Numerous studies have been carried out in the areas of foundation and structural engineering to determine the values of surface deformations admissible for different types of structures. These values depend on the type of structures, their condition and their operational requirements. They are usually expressed in terms of a slope of a non-uniform settlement profile rather than in absolute values. In the past, the theory of foundation engineering used to recognize two limit states of a settlement profile slope:

a) the structural limit state, when the structural integrity of a building is compromised ;
b) the operational limit state, when some essential functions of building are at risk.

More recently, due to the decreasing public tolerance to large projects and due to increasing concern for environmental protection, a third limit state is often considered for tunnel projects:

c) the cracking limit state, when visible cracks appear, but no operational or other essential functions of the building are affected.

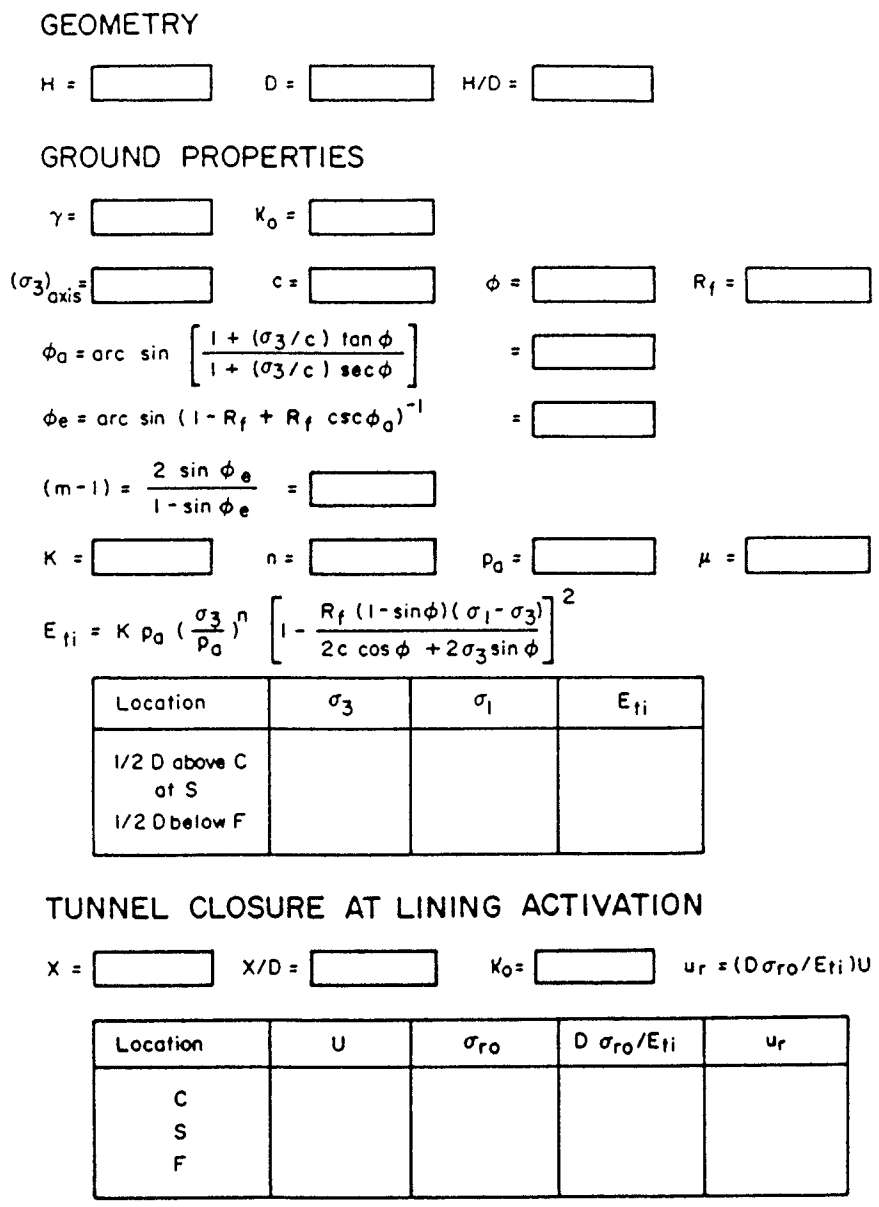


Figure 5. Calculation sheet with geometric ground properties data and tunnel closure at lining activation.

REFERENCE VALUES

H/D= [ ] K_o= [ ] (m-1)= [ ]

	$1-\Sigma_{ref}=\alpha_{ref}$	α_{ref}/U_{ref}	U_{ref}
C			
S			
F			

STRESS RELEASE AND STIFFNESS CHANGE AT LINING ACTIVATION

	U	U/U_{ref}	Beyond Limit?	λ	λ'
C					
S					
F					

STRESSES : $\Sigma = 1-(1-\lambda)\alpha_{ref}$

	$(1-\lambda)$	Σ
C		
S		
F		

$\overline{\Sigma}=(\Sigma_C+2\Sigma_S+\Sigma_F)/4$ = [ ]

$\overline{\alpha} = 1-\overline{\Sigma}$ = [ ]

$\gamma_{red} = \gamma \cdot \overline{\Sigma}$ = [ ]

STIFFNESS : $E_t = (\lambda'/\lambda'_i)\,E_{ti}$

	λ'_i	E_{ti}/λ'_i	E_t
C			
S			
F			

$\overline{E}_t = (E_{tC}+2E_{tS}+E_{tF})/4$ = [ ]

Figure 6. Calculation sheet for estimates of the stress release and ground stiffness at the section the lining is activated.

Of the three limit states it used to be usually the second one which was applied to design of a tunnel. Specific values of limit state settlement profile slope are given in most standards or codes of practice of foundation engineering and will not be discussed or reviewed here. In this context, it is important to realize that these values are of a semi-empirical nature, derived from field measurements and supported by theoretical analysis. Irrespective of their sources (USA, Great Britain, Russia and others) the standardized values of surface settlement limit slopes differ very little, which supports their credibility.

2D STABILITY VERIFICATION

$\bar{\alpha}$ = [ ] $\bar{\sigma}_{ro} = \frac{1+K_o}{2}\gamma(H+\frac{D}{2})$ = [ ] ϕ = [ ] c/γD = [ ] H/D = [ ]

$\therefore P_\ell/\gamma D$ = [ ] P_ℓ = [ ] $\alpha_\ell = 1 - P_\ell/\bar{\sigma}_{ro}$ = [ ]

$$FS = \frac{[(1-K_o)/(1+K_o)+\alpha_\ell]}{[(1-K_o)/(1+K_o)+\bar{\alpha}]} = [\ ]$$

LINING-GROUND INTERACTION

GEOMETRY : Z_o = [ ] r_o = [ ] r_i = [ ]

SOIL : $\bar{E}_t$ = [ ] μ = [ ] γ_{red} = [ ] K_o = [ ]

LINING : E_s = [ ] μ_s = [ ] d = [ ] I_s = [ ]

ITERATION No.: _____ $\bar{E}_t$ = [ ] $\dot{\alpha}$ = [ ] β = [ ]

	u_r	Δu_r	$u_{rf} = u_r + \Delta u_r + v''_{zo}$	U	U/U$_{ref}$	λ'	E_t	σ_r
C								
S								
F								

$v''_{zo} = \frac{(1+\mu)(3-4\mu)}{4(1-\mu)\bar{E}_t}\gamma_{red}\cdot r_o^2 \ell n \frac{r_o}{r_i}$ = [ ] $(\bar{E}_t)_{new} = (E_{tC} + 2E_{tS} + E_{tF})/4$ = [ ]

Figure 7. Calculation sheet for 2D stability verification and lining–ground interaction analysis.

ITERATION No.: _____ $\bar{E}_t$ = [ ] $\dot{\alpha}$ = [ ] β = [ ]

	u_r	Δu_r	$u_{rf} = u_r + \Delta u_r + v''_{zo}$	U	U/U$_{ref}$	λ'	E_t	σ_r
C								
S								
F								

v''_{zo} = [ ] $(\bar{E}_t)_{new}$ = [ ]

ITERATION No.: _____ $\bar{E}_t$ = [ ] $\dot{\alpha}$ = [ ] β = [ ]

	u_r	Δu_r	$u_{rf} = u_r + \Delta u_r + v''_{zo}$	U	U/U$_{ref}$	λ'	E_t	σ_r
C								
S								
F								

v''_{zo} = [ ] $(\bar{E}_t)_{new}$ = [ ]

Figure 8. Calculation sheet for interactive analysis of the lining–ground interaction.

SUBSURFACE SETTLEMENTS

CROWN SETTL. = [ ] U/U_{ref} = [ ] λ_c = [ ] $\alpha_c = (1-\lambda_c)\alpha_{ref}$ = [ ]

K_o = [ ]

ϕ = [ ]

DEPTH Z (m)	Z / H	SETTLEMENT/CROWN SETTL. H/D=	H/D=	H/D=	SETTLEMENT
H=	1.000	1.000	1.000	1.000	

SURFACE SETTLEMENTS

α_c = [ ]

K_o = [ ]

ϕ = [ ]

DIST. Y (m)	0.0	
Y/D	0.0	
SETTL./CROWN SETTL. H/D=		MAX. DISTOR.:
H/D=		1: [ ]
H/D=		AT: [ ]
SETTL.		

Figure 9. Calculation sheet for subsurface and surface settlements.

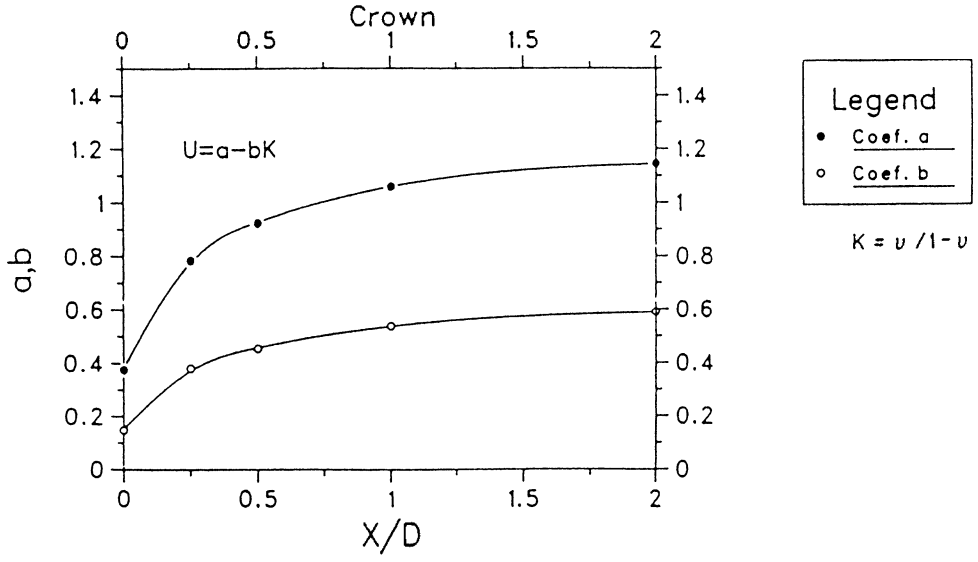


Figure 10. Coefficients of the U and K relationship for the crown of an unlined shallow tunnel (1.5<H/D<3.0), as a function of the distance to the face.

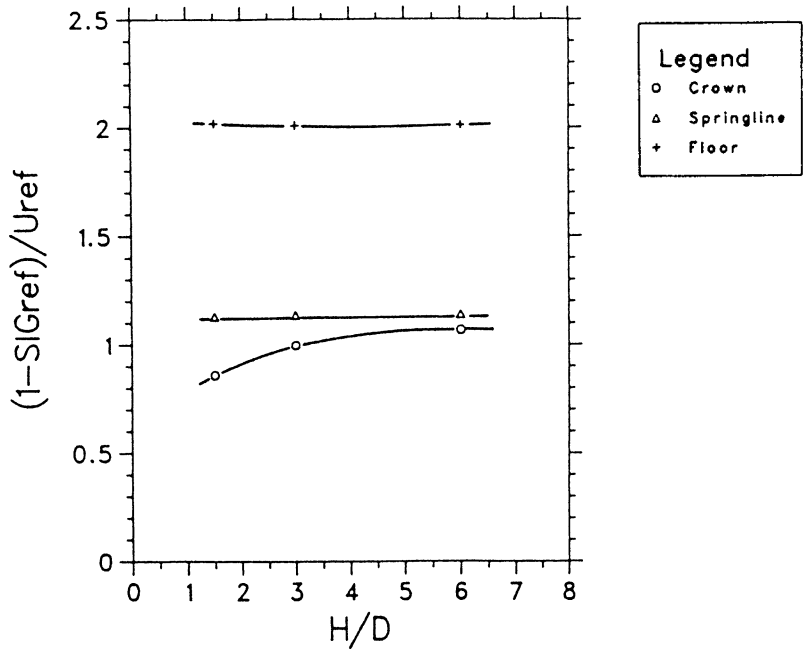


Figure 11. Slope of the arbitrary axes used to obtain the NNGRC represented as a function of H/D, for K_o=0.8.

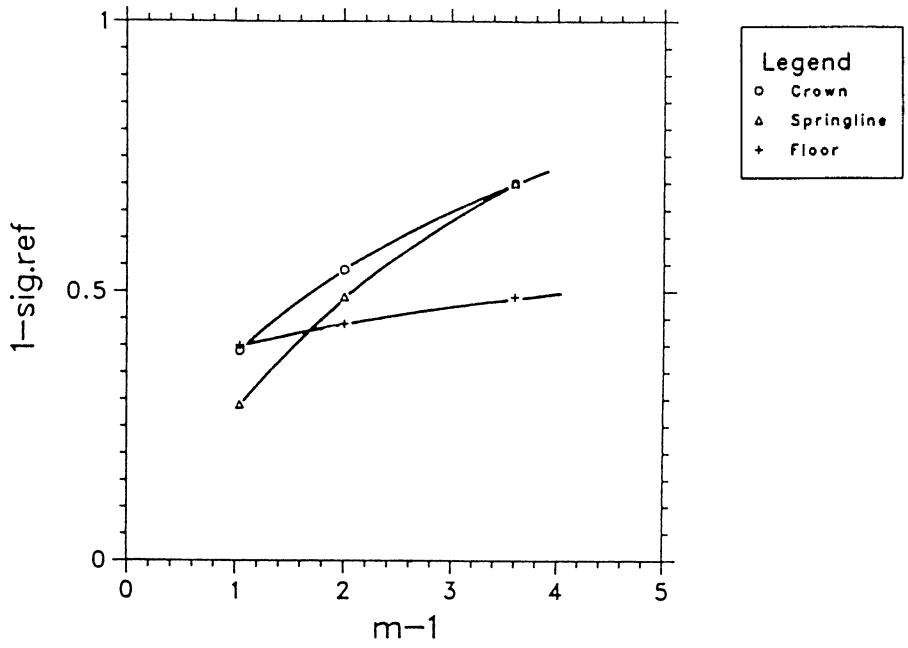


Figure 12. Relationships between a_{ref} and the friction angle for K_o=0.8.

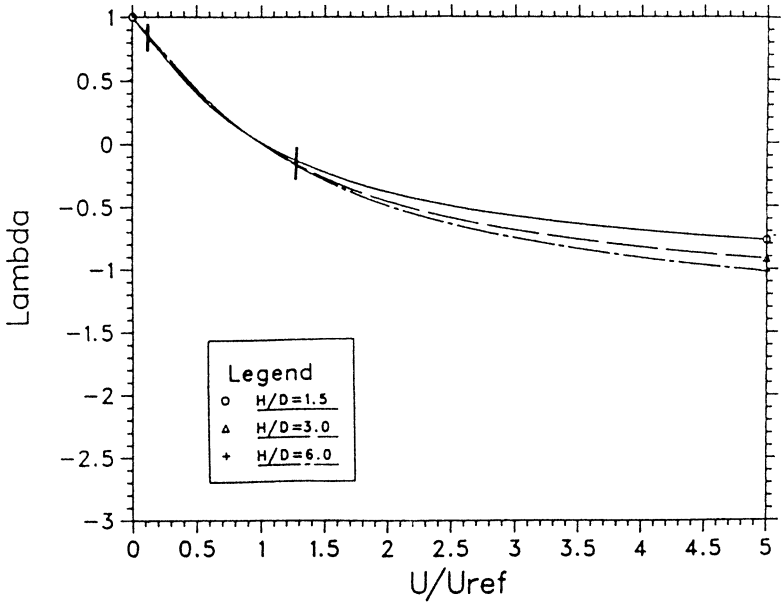


Figure 13. NNGRCs for tunnel crown, calculated for K_o=0.8.

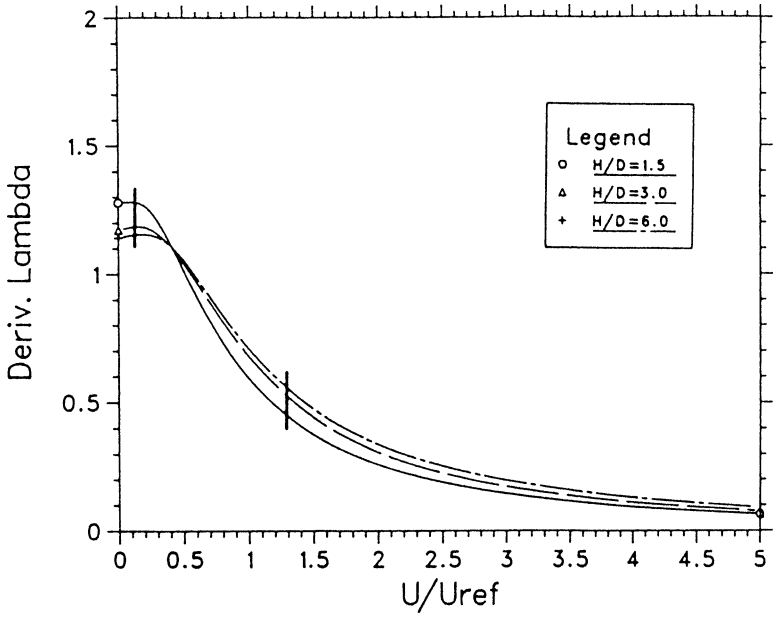


Figure 14. Variations of λ' for the tunnel crown, calculated for K_o=0.8.

Nevertheless, public and political pressures more and more force tunnel designers to apply limit state c). To avoid any visible cracks or, in different words, to excavate a tunnel under buildings completely unnoticed, requires to achieve a surface settlement trough in the order of 1:1000 or flatter. While this is possible in principle, it requires substantial extra cost, far in excess of the cost of crack repair or even some compensation to the building owners. The introduction and acceptance of and the third (sub c)) limit state represents a major burden on

urban tunnelling and as such must be dealt with the international tunnelling community with some urgency.

4 GROUND RESPONSE TO TUNNELLING

As discussed above, the design of urban tunnels requires, among others, the prediction of the magnitude and distribution of the loads on the lining and of the displacements throughout the surrounding ground, including settlements at the ground surface.

Several methods have been proposed to predict the lining load and ground displacements. However, most of them deal with these two predictions separately, using different semi-empirical and/or analytical methods, both for the settlement and lining predictions. The separation of the problem of displacement from the problem of lining pressure is convenient analytically, but is entirely artificial. In reality, soil displacements are closely linked to soil pressures. Although this interrelation is well known through the convergence-confinement concept (Figure 1) its full appreciation in the practice of shallow tunnel design is not always apparent. As a consequence, many design approaches incorporate inconsistent and even conflicting assumptions, which simultaneously lead to higher lining loads and larger settlements than actually observed.

By recognizing these limitations in current practice a new approach to the problem has been attempted, which accommodates the dependency between stresses and displacements (Eisenstein & Negro 1985, Negro 1988). This new approach is based on the concept of ground and support interaction. Ground response curves are obtained by numerical modelling for distinct points around the tunnel perimeter and for varying ground and geometric conditions. The effect of delayed lining installation is calculated, followed by closed form solution for the lining-ground interaction. Equilibrium stresses and displacements at the tunnel opening are thus defined. The knowledge of these displacements allows the magnitude and distribution of the settlement above the tunnel to be estimated. In this way, lining loads and ground displacements are obtained simultaneously as a result of a single analysis.

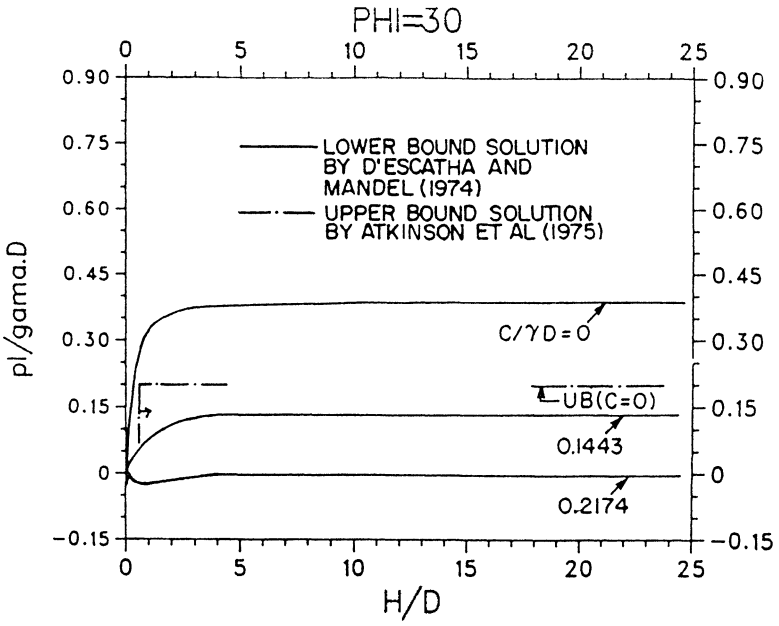


Figure 15. Lower and upper bound solutions for tunnel collapse pressure in a frictional and cohesive soil (ϕ=30°).

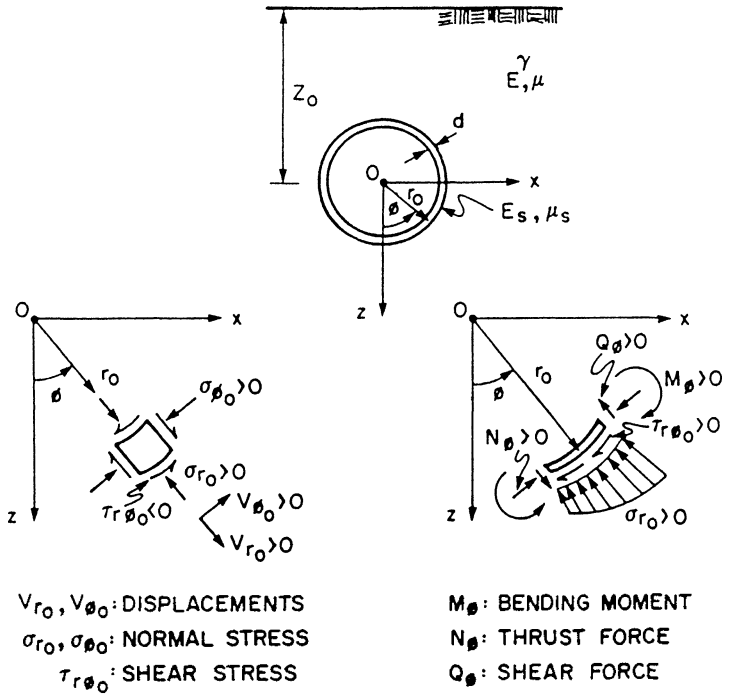


Figure 16. Notations and conventions used in Hartmann's solution.

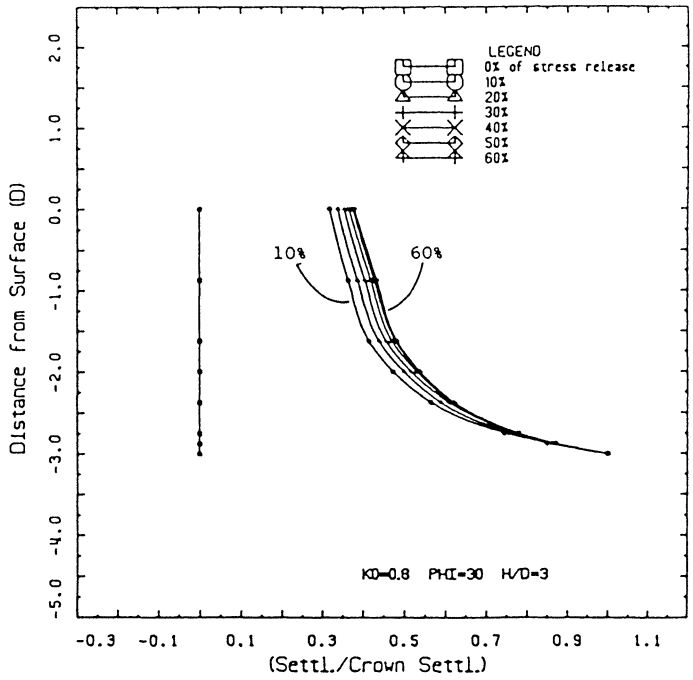


Figure 17. Normalised subsurface settlements for H/D=3.0, K_o=0.8 and ϕ=30°, calculated for increasing amount of stress release.

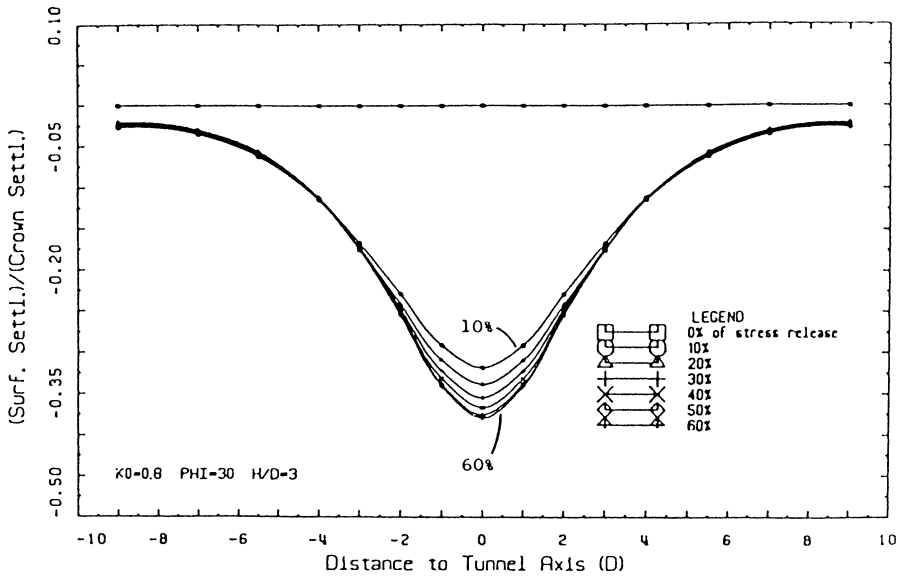


Figure 18. Normalised surface settlements for H/D=3.0, K_o=0.8 and ϕ=30°, calculated for increasing amount of stress release.

The analysis assumed the ground behaviour to be non-linear and time independent and the tunnel to be shallow, single and circular. The numerical modelling was developed through generalization of results obtained by two and three dimensional finite element analyses.

The two dimensional modelling allows relationships to be established between tunnel closure, ground mass displacements, the amount of ground stress release and the resulting ground stiffness changes. The results of the generalization process are presented in dimensionless form, through equations and charts, which allow the above relationships to be defined for distinct soil strength, ground stiffness, in-situ conditions and for different tunnel sizes and depths.

The effects of the delayed installation of the lining are accounted for by combining the above results with an estimate for tunnel closures, developed from three dimensional finite element parametric analyses. The lining-ground interaction is taken into consideration through an analytical closed form solution which includes the effects of the gravitational stress field.

The development of the design procedure was limited to circular tunnels, with full face excavation under time independent conditions, in situations not involving ground failure, shear dilatancy and post peak softening. However, when the method was verified against a large number of actual case histories, some of which included deviations from the defined conditions, reasonable results were still obtained for both ground and lining response. In each of the cases investigated, good ground control conditions existed and approximations were introduced to account for features such as tunnelling below the water table with partial consolidation of the soil mass, construction involving the use of compressed air or positive face control, staged excavation of the tunnel face with a temporary invert in the heading excavation (NATM, SEM or SM methods), non circular tunnel profiles or mixed face conditions.

The generalized procedure proposed by this approach provides a coupled solution for the ground response and while making an extensive use of finite element studies does not actually requires any finite element analysis to be performed. This is because the results were eventually processed and reduced by dimensionless analysis to produce charts and equations, directly usable in design.

To facilitate this dimensionless data reduction and to allow generalization, one particular family of hyperbolic functions has been selected to represent the ground, those that exhibit homologous stress-strain curves, because these can be normalized into one single dimesionless relationship. Figure 2 presents one set of curves that illustrates these properties.

Table 1 Tunnelling methods for urban tunnels in soft ground

Tunnelling method and ground conditioning	Total number of cases	Type of ground		
		Granular	Fine-grained	Mixed
NATM, SEM, SM (all)	7	5	0	2
NATM, SEM, SM (with ground conditioning)	5	3	0	2
NATM, SEM, SM (without ground conditioning)	2	2	0	0
OF TBM (all)	12	2	1	9
OF TBM (with ground conditioning)	8	1	0	7
OF TBM (without ground conditioning)	4	1	1	2
CF TBM (all)	4	0	2	2
EPB TBM (all)	20	6	1	13
SPB TBM (all)	25	13	2	10
APB TBM (all)	5	3	1	1
Others	1	0	0	1
TBM (all types)	66	24	7	35
TOTAL	74	29	7	38

Legend:
TBM – Tunnelling Boring Machine
OF TBM – Open Face TBM
CF TBM – Closed Face TBM
APB TBM – Air Pressure Balance TBM
SPB TBM – Slurry Pressure Balance TBM
EPB TBM – Earth Pressure Balance TBM

Table 2 Tunnelling methods in urban environments by percentage distribution

Tunnelling method	% of all
TBM (all types)	89
SPB TBM	34
EPB TBM	27
OF TBM	15
APB TBM	7
CF TBM	6
NATM	9
Others	1
TOTAL	100

Legend:
TBM – Tunnelling Boring Machine
OF TBM – Open Face TBM
CF TBM – Closed Face TBM
APB TBM – Air Pressure Balance TBM
SPB TBM – Slurry Pressure Balance TBM
EPB TBM – Earth Pressure Balance TBM

The ground response curve (GRC) is defined as the relationship between the radial stress at the tunnel perimeter and the corresponding radial displacement. Unlike the convergence curves used in the convergence-confinement method, the GRC is different for each point of the tunnel perimeter. This difference is due to the presence of the free boundary surface, to gravity and to Ko values differing from unity. The ground response curves can be easily obtained by the FEM model described. They can be then conveniently plotted in dimensionless form as Normalized Ground response Curves (NGRC), an example of which is shown on Figure 3. At least three

NGRC's are needed (F for the floor, S for the springline and C for the crown) for one tunnel.

Since the stress-strain curves for different stress levels are of a homologous nature, it can be further shown that the NCRC in terms of normalized, dimensionless stress and displacement are unique for a given point at the tunnel perimeter, for a given ratio of overburden cover H to tunnel diameter D (H/D) and for any constant value of Ko. Obviously, different NGRC's are obtained for different ratios H/D, different soil strengths and different Ko's. These curves have been found to be relatively insensitive for the usual range of values of Poisson's ratio.

The proposed method is useful for basic design including alignment optimization, selection of construction procedures, the influence of parameter variability, etc. The simplicity of the procedure permits ease in design reevaluation during tunnel construction as part of an observational design approach. The method can also be applied and calibrated with field monitoring.

The analysis can be either undrained or drained and should be assessed according to the ground profile, soil type, its properties and the construction scheme to be used.

The guidelines provided below were developed for a cohesionless soil model, but the design approach can be easily adapted for a frictionless soil model which would better describe the behaviour of saturated soils under undrained soil conditions. Figure 4 shows a flow chart which summarizes the main steps of the procedure. The set of calculation sheets (Figures 5 to 9) facilitates the calculations involved. The steps include:

a) The assessment of the geometric conditions of the tunnel (Figure 5).
b) The evaluation of the geological and geotechnical aspects involved, including the definition of ground parameters (Figure 5).
c) The assessment of the construction technology to be applied, including details of the support system.
d) The assessment of the applicability of the proposed design procedure, including the anticipation of likely ground conditions to be encountered.
e) The evaluation of the pre-support ground response (tunnel closure, amount of stress release, ground stiffness change, ground stability verification (Figures 5, 6 and 7).

The amount of tunnel closure requires definition of the radial displacements which have occurred up to the instant the support is activated. An estimation of the distance (x) from the face to the location of support activation enables the dimensionless radial displacements (U) for the crown, springline and floor to be established (from Figure 10). The procedure developed to estimate the amount of tunnel closure at the section where the support is activated was established through parametric three-dimensional finite element analysis. It is an approximate solution assuming an elastic medium, but provides sensible estimation of radial displacements at the three points of the tunnel contour for different cross sections at or behind the face. From case histories, it is seen that good ground control conditions are generally associated with U values less than 1.0, and for collapse conditions, U is about 1.8.

These three dimensional analyses also allowed simplified solutions to be obtained for estimates of the maximum horizontal displacement at the tunnel face and of the maximum longitudinal distortions for the ground surface and subsurface ahead of the face. These are important for assessing the stability of the face and heading.

Reference values (Figure 6) for the amount of stress release (α_{ref}) and for the dimensionless displacements at the tunnel contour (U_{ref}) are found for the particular geometry (H/D), the in-situ stress ratio (K_o) and for the adjusted friction angle (m-1). This requires, for example, Figures 11 and 12 to obtain α_{ref}/U_{ref} and α_{ref}, from which U_{ref} is calculated for the crown, springline and floor.

Defining

$$\lambda = 1 - \alpha / \alpha_{ref} \tag{1}$$

and

$$\lambda' = \delta\lambda / (\delta U/U_{ref}) = (U_{ref}/\alpha_{ref}) (D / E_{ti}) (\delta\sigma_r/\delta u_r) \tag{2}$$

one can plot these two parameters as a function of U/U_{ref}. The plots for different H/D ratios fit into a very narrow band, allowing a second normalization of the ground reaction curve (GRC).

The twice normalized ground reaction curves (NNGRC or λ), and their derivative function curves (λ'), shown for example in Figures 13 and 14 respectively, permit the calculation of the corresponding stress (Σ) and the current ground moduli (E_t) at the three contour locations. Note that λ curves are related to the amount of ground stress relaxation and the slopes of these curves (λ curves) provide the ground stiffness for any amount of tunnel closure.

The reduced unit weight of the soil and reduced ground stiffness are found by averaging the stress ratio (Σ) and and the current stiffness (E_t) according to the equations provided on the calculation sheet in Figure 6.

The verification of tunnel stability is usually limited to the stability of the unsupported excavation of the heading and face. The problem is three dimensional and its solution involves considerable approximation. Generally the 3D stability of the ground is controlled by the geometric conditions and soil strength. Available plasticity solutions perhaps provide the most appropriate estimates for the design stage.

During construction, the assessment of global stability problems may be effectively done by calculating the longitudinal distortions from the longitudinal distribution of ground settlements. A check of the two dimensional tunnel stability is made taking into account the reduction of the ground stress calculated at the point the support is to be activated. This is done using, for example, the lower bound solutions presented in Figure 15.

The minimum acceptable factor of safety is defined in terms of the maximum tolerable ground deformations and is calculated according to the expression given in Figure 7. Typically, factors of safety used in soil tunnelling are low and stability deteriorates rapidly with increasing ground displacements. However, if the calculated factors of safety are acceptable (greater than, say, 1.3 to 1.6) the construction procedure is applicable and the proposed design method can be used.

f) The analysis of the lining-ground interaction including the calculation of thrust forces, bending moments and lining construction (Figures 7 and 8).

A two dimensional linear elastic analytical model was selected to model the lining-ground interaction phase of the shallow tunnel design. In many situations, the assumption of linear elasticity is reasonable at this stage of the ground convergence curve.

g) The prediction of the ground movements, including final tunnel closure, subsurface and surface settlements and maximum ground distortions Figure 9).

All of these predictions are related to the crown displacements occurring. The total crown displacement at equilibrium (u_{rf}) if used to determine the associated amount of stress release (α_c) using the λ curves for the tunnel crown (e.g. Figure 13). Subsurface and surface ground settlements are calculated from charts such as in Figure 17 and 18 and may require interpolation. Parametric finite element studies revealed that the unlined tunnel analyses, with a crown displacement equal to the equilibrium crown settlement calculated in the lined tunnel case, furnished settlement distributions similar, yet slightly conservative, to those found in the latter case.

It is sufficient, therefore, to calculate the amount of stress release in the unlined tunnel solution causing the equilibrium crown displacement obtained in the lining-ground interaction analysis.

Both the maximum ground distortion and the width of the surface settlement trough as estimated by the calculation procedure are typically unconservative when compared with field measurements. This trend was not unexpected because of the nature of the finite element modelling.

An empirical correction factor was introduced for the calculated maximum distortion (λ_c) such that the corrected distortion would be $\lambda_{cc} = 1.4\ \lambda_c$.

A semi-empirical correction is applied for the width of the settlement trough (w_s) using the maximum surface settlement (s_{max}) as calculated by the design method, and γ_{cc} as given above.

The shape of the settlement trough is assumed to follow a normal probability curve giving:

$$w_s = 1.52\, s_{max} / \gamma_{cc} \quad (3)$$

The proposed method was verified against a large number of actual case histories. These always involved good ground control conditions, reflecting either good construction quality or simply good ground quality. In order to test the performance of the method in actual practice, some of the case histories included conditions that deviated considerably from those originally set up in the development of the method.

Despite this, the calculation method yielded sensible results, both in terms of ground and lining responses. Noted discrepancies were duly explained in terms of the limitations of the method. The results obtained are summarized in Negro (1988). The exercise covered shielded and mined tunnels, different support systems and lining stiffnesses, a wide range of geometric conditions and a fairly wide range of soil types. This study revealed that for a quick lining-ground interaction analysis, it may be sufficient to assume a 50% reduction of the in-situ ground stresses and of the in-situ tangent ground stiffness, provided the ground control conditions are comparable to those found in the cases studied.

It was suggested that the proposed calculation method seems to provide reasonable predictions of tunnel performances, whenever the factor of safety of the ground at the section the support is activated, is within or above the calculated range of factors found for these cases (from 1.2 to 2.0). The final dimensionless crown displacement U was found to vary between 0.5 and 1.0, and quick conservative estimates of the final crown displacement could therefore be performed by making U=1.0.

Full range of charts and diagrams facilitating the usage of this method in a variety of geometric and geotechnical conditions ca be found in Negro (1988).

5 DEVELOPMENT OF OBSERVATIONAL METHOD

The Observational Method (OM) is eminently well suited for tunnelling. Tunnels are line structures, allowing for modifications in design and construction during their excavation.

The progress of the OM has been made mainly due to rapid advancements in the areas of electronics, data processing and measuring techniques in general. The developments in these areas were then transferred to tunnelling.

The Observational Method in tunnelling usually follows some or all of these goals:

a) Verification of design – where design assumptions are checked and modified during excavation of a tunnel, with positive consequences for safety and economy of the project.
b) Quality control of construction – as a method of control of contract specifications.
c) Warning system against major failures or collapses – as such events seldom come without some prior indications, the OM can serve as a warning system providing the monitoring data are continuously interpreted.

6 GROUND CONDITIONING

In tunnelling, the selection of an appropriate tunnelling method usually involves one of these two principal approaches:

a) either a tunnelling method is selected, which is suitable for and capable of controlling the given ground, or
b) the given ground is conditioned to be made suitable for and controllable by the selected tunnelling method.

The first approach (sub a)) will be discussed in the next section of this paper. It is the second (sub b)), which leads to ground conditioning. Major advances in this category included:

1) grouting of soil or rock mass voids in order to improve ground properies (formation grouting);
2) jet grouting to improve properties of ground with voids difficult to be penetrated by grout under pressure;
3) compensation grouting to compensate for loss of ground due to tunnelling and to control surface settlement;
4) compaction grouting to prestress the ground prior to destressing due to tunnel excavation.
5) freezing, as a temporary measure to stabilize otherwise unstable, water saturated ground
6) ground water lowering.

The progress in the above categories has not been uniform. Major advances were recorded in categories 1. to 4. However, these methods of ground conditioning depend to a large degree on skills and experience of their providers and on quality of the work. Furthermore, these methods, so far, resisted any development of a scientific background and thus still remain largely empirical.

Another aspect which makes these methods somewhat unreliable is the fact, that it is difficult to control the quality and consistency of their application. Consequently, there is a demonstrable trend in tunnelling to substitute ground conditioning by using tunnel boring machines with positive face control. This technology will be discussed in the next section.

7 TUNNEL BORING MACHINES WITH POSITIVE FACE CONTROL

The ultimate goal of mechanized tunnelling would be a TBM capable of handling any type of ground. While fulfilling such a dream is still somewhere in the future, the development of tunnel boring machines with positive face control has been a major step in that direction.

The present state of application of tunnelling technology for shallow urban tunnels in soft ground has been clearly dominated by TBM with positive face control. This is clearly demonstrated on Tables 1 and 2, showing the results of a major survey of urban tunnelling worldwide, involving 74 case histories of shallow tunnels excavated during the last 20 years in 18 countries (Eisenstein et al. 1997). The tunnel boring machine category includes three types of TBM – earth pressure balance (EPB) TBM, bentonite slurry (BS) TBM and a TBM with compressed air (CA). The third (CA) used to be the only positive face control technology available prior to the development of the EPB and BS techniques, however presently is gradually getting extinct in tunnelling practice because of difficult working conditions associated with its application. Thus it is the first two types (EPB and BS) which convincingly prevail in soft ground urban tunnelling. Both types are clearly preferred over open face TBM and nonshielded methods such as the New Austrian Tunnelling Method, because they offer far better ground control. In urban environment ground control is of paramount importance, since it directly affects surface settlement and as such often represents the governing design and construction criterion.

While both the EPB and BS TBM can deliver a positive face control during tunnel excavation, they differ fundamentally in the way this control is provided. The EPB TBM pressurizes the face of the excavated tunnel by stresses transmitted by the reworked soil cuttings present in the front chamber. The face support is thus of the earth pressure type and as such is deformation dependent. Because the soil cuttings are removed and the soil pressure is released mechanically through a screw conveyor, the EPB can only work in either a total stress or an effective (grain to grain) stress mode. It cannot support neutral (water) pressure inside the front chamber (plenum).

The BS TBM provides stress at the face hydraulically by bentonite slurry kept under pressure in the front chamber. The slurry penetrates some distance into the supported soil and creates a membrane against which the pressure is maintained. Thus the BS TBM can control either a total stress or a neutral (water) pressure. It cannot support directly the effective stress in the chamber.

This difference in the mechanics of the face support had a direct impact on the applicability of the two TBM types. The EPB TBM can successfully control and support a tunnel face in either dry or a saturated fine grained soil, where no free water is present in the front chamber.

On the other hand, the BS TBM can reliably operate in practically all types of soils, fine or coarse grained, with or without free water. This is because free water can be effectively countered by the pressurized bentonite slurry.

Both types, when applied in proper ground conditions, can deliver a ground control superior to most other tunnelling technologies. This explains, why TBM with positive face control presently account for majority of urban tunnel construction.

8 SUMMARY

Major progress has been achieved in design and construction of shallow urban tunnels. New theoretical models have been developed to predict ground behaviour in general and surface settlement in particular. Most important progress, however, has been recorded in the development of tunnel boring machines with positive face control, such as the earth pressure balance and the slurry TBM. Because of their superior ability to control the ground, the machines currently dominate the practice of urban tunnelling.

REFERENCES

Eisenstein, Z. & A. Negro 1985. Comprehensive design method for shallow tunnels. *Proceedings ITA/AITES International Conference "Underground Structures in Urban Areas"*, Prague, Sept. 1985, 3-20.

Eisenstein, Z. Martin, G.R. & H.W. Parker 1997. The Challenges of Tunnelling for Los Angeles Subway. *Proceedings ITA/AITES International Conference "Underground Construcion '97"*,Prague, Oct. 1997, 107-122.

Negro, A. 1988. Design of Shallow Tunnels in Soft Ground. *Ph.D. thesis, University of Alberta, Edmonton, 1480 pp.*